



Pascal Triangle Row Generation

Brief on Pascal's Triangle

Pascal's Triangle is a triangular array of numbers used to represent binomial coefficients. The triangle starts with a single 1 at the top. Each number in the triangle is the sum of the two numbers directly above it in the previous row. The construction rules are as follows:

- The first and last elements of each row are always 1.
- Any other element at row n and position k (zero-indexed) is given by the binomial coefficient:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

The first few rows of Pascal's Triangle are:

Row 0:						1					
Row 1:					1		1				
Row 2:				1		2		1			
Row 3:			1		3		3		1		
Row 4:		1		4		6		4		1	

Each row also corresponds to the coefficients of the binomial expansion of $(a + b)^n$.

Problem Statement

Design a digital system that meets the following specifications:

1. Accept an input integer n ($n > 2$) through the Switch and generate the n^{th} row of **Pascal's Triangle**.
2. Display each element of the row on 8 LEDs using its **8-bit binary representation**.
3. Introduce a delay of 2 seconds between displaying each number.
4. Use a **50 MHz clock** to generate the 2-second delay.

The entire system should be implemented by designing a **Finite State Machine (FSM)** using **VHDL behavioral modeling**.

Example Output (for $n = 3$)

The binary 8-bit representation for row 3 is:

00000001 (1), 00000011 (3), 00000011 (3), 00000001 (1)

Each number should be displayed on LEDs with a delay of 2 seconds between them.

Constraints

- $n > 2$
- Delay between each LED pattern is 2 seconds